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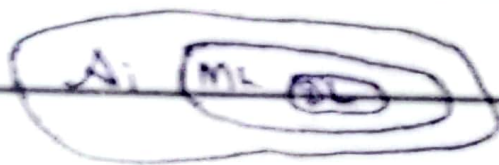
Deep Learning

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What is deep learning?

Deep learning is a subfield of AI and ML that is inspired by the structure of a human brain.

- D.L algorithms attempt to draw similar conclusions as humans would by continually analyzing data with a given logical structure called neural network



ML algorithms mostly focuses on

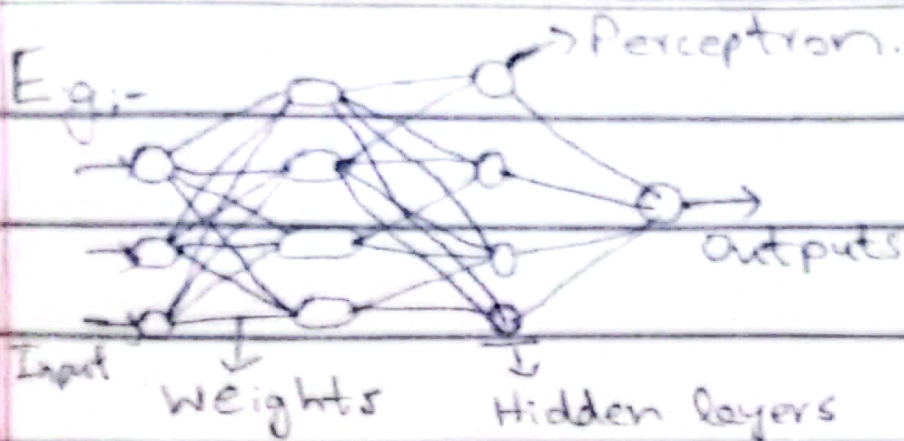
- statistical techniques, Whereas

D.L depends on a logical structure called neural network.

Basically a logical structure $\swarrow$ $\searrow$ Inspired by human brain.

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Name d.L is connected with these hidden layers that's why name DL

- ANN
- RNN (Recurrent neural network) $\rightarrow$ Speech/text.
- CNN [convolutional " " $\rightarrow$ Image
- GAN $\rightarrow$ generate things.

Why d.L so famous:-

- 1) Applicability $\rightarrow$ Applied to wide domain of problem
- 2) Performance

$\rightarrow$ DL is a part of ML methods based on a.NN with representation learning

P.T.O. $\rightarrow$

Less data $\rightarrow$ ML wins

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Representation Learning:

Automatic feature extraction

- Feature engineering isn't needed in DL
e.g. In ML Dog/cat class. for
we featured on size, colour etc.

DL VS ML:-

- 1) Data dependency:- Data $\uparrow$ hungry
- 2) Hardware dependency:- $\rightarrow$ costly in DL.
 $\rightarrow$ Matrix multiplication $\rightarrow$ GPU
 $\downarrow$
Powerful
- 3) Training time:-
T.T is very high in D.L.
- 4) Feature selection:-
- 5) Inter-pretability:- Quite low & its flaw. (Black box)
In DL models its $\downarrow$ means
we can't provide solid proof that
why u ain't selected or why u learned.